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(Telangana State Private Universities Act No. 13 of 2020 &
G.O.Ms. No. 14, Higher Education (UE) Department)

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Department of Information Technology

III B.Tech I Sem - AD

Industry Oriented Internship – Deep Learning Techniques Development

MR24-1IT0163

Financial Document Tampering Detection System Using EfficientNetV2B0 and Grad-CAM

Batch no. 13

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Guide

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Table of Contents

- Introduction
- Literature Survey
- Limitations of Existing Algorithms/Methodologies
- Abstract
- Proposed System and Methodology
- Hardware/Software Requirements
- Results on the Test Receipts
- From 97.85% to a Working Detector
- Screenshots of the Application
- Conclusion and Future Scope

Introduction:

Financial document fraud has become increasingly common with the widespread use of digital editing tools. Documents such as invoices, receipts, bank statements, and bills can be manipulated to alter financial information, resulting in significant losses for organizations. Traditional manual verification methods are time-consuming and prone to human error. This project proposes an Explainable AI-based system that automatically detects whether a financial document is genuine or tampered. The system uses EfficientNetV2B0 for document classification and Grad-CAM to highlight suspicious regions, making the prediction transparent and easy to understand.

Literature Survey:

Several researchers have applied deep learning techniques to document forgery detection using Convolutional Neural Networks (CNNs), EfficientNet, and Vision Transformers. These methods have shown improved accuracy compared to traditional image processing approaches. Explainable AI techniques such as Grad-CAM have also been introduced to visualize the regions responsible for model predictions. However, many existing systems focus on document classification without providing clear visual explanations, making their decisions difficult to interpret.

Limitations of Existing Algorithms/Methodologies:

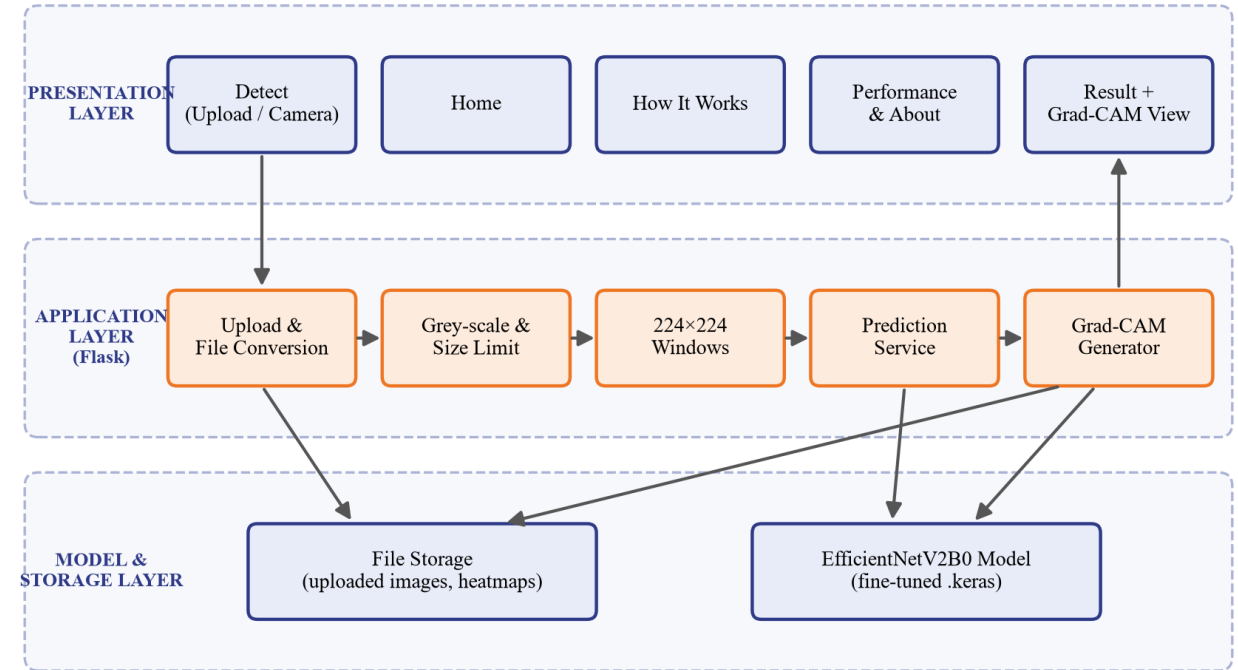
- Manual verification is slow and depends on human expertise.
- Traditional image processing techniques cannot detect complex digital manipulations.
- Many deep learning models act as black boxes and do not explain their predictions.
- Existing systems often fail to generalize across different types of financial documents.
- Several methods require expensive computational resources or specialized datasets.

Abstract:

Financial document tampering is a serious problem for banks, insurance companies and businesses. FinDoc AI detects tampered receipts and invoices and explains every decision. It is trained and evaluated on the public “Find it again!” dataset of 988 scanned receipts. Error Level Analysis (ELA) re-saves the page as JPEG and finds text regions whose compression error differs from the rest, which is typical of pasted or re-typed content. EfficientNetV2B0, fine-tuned on ELA maps, classifies the most suspicious regions, and Grad-CAM heatmaps with plain-language reasons explain the result. Typed Word, PowerPoint and digital PDF files are checked through their metadata. The Flask web application is deployed on Vercel and accepts images, camera photos, PDF, Word and PowerPoint files. On 218 test receipts the system reaches 71.10% accuracy and a document-level AUC of 79.16%, detecting 23 of 35 tampered receipts.

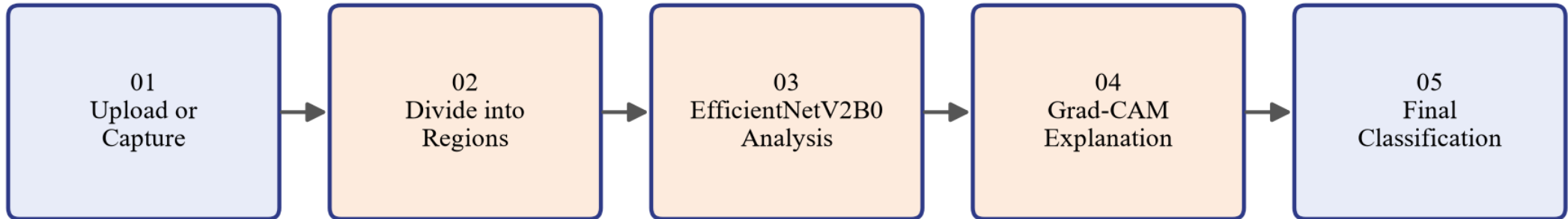
Proposed System:

- Upload an image, PDF, Word or PowerPoint file, or capture with the camera
- Scanned pages: Error Level Analysis, re-saved as JPEG (quality 70) and compared
- Score = how far the most unusual 64x64 text region differs (z-score); EfficientNetV2B0-ELA checks it
- Grad-CAM heatmap, flagged regions and plain-language reasons
- Digital files: metadata check (editor, dates, re-saves) instead of pixels



Methodology:

ELA re-saves the page as JPEG and measures the re-compression error of every text pixel; edited areas were compressed a different number of times and stand out. EfficientNetV2B0 (ImageNet weights) is fine-tuned on 4,333 ELA patches (1,638 tampered), reaches 80.4% patch AUC, and runs in the web app with ONNX Runtime.



Regional scores are combined into a document-level decision of Genuine or Tampered

Hardware Requirements:

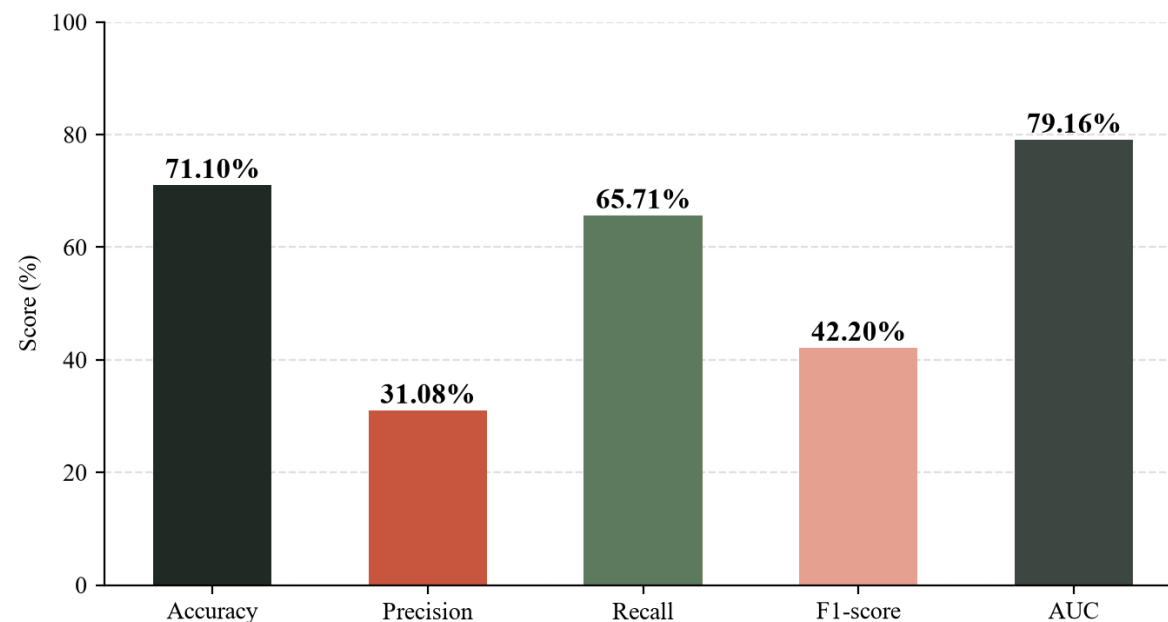
- Processor: Intel Core i5 or above
- RAM: 8 GB (16 GB for training)
- Storage: 10 GB free space
- Camera (optional, for capturing documents)

Software Requirements:

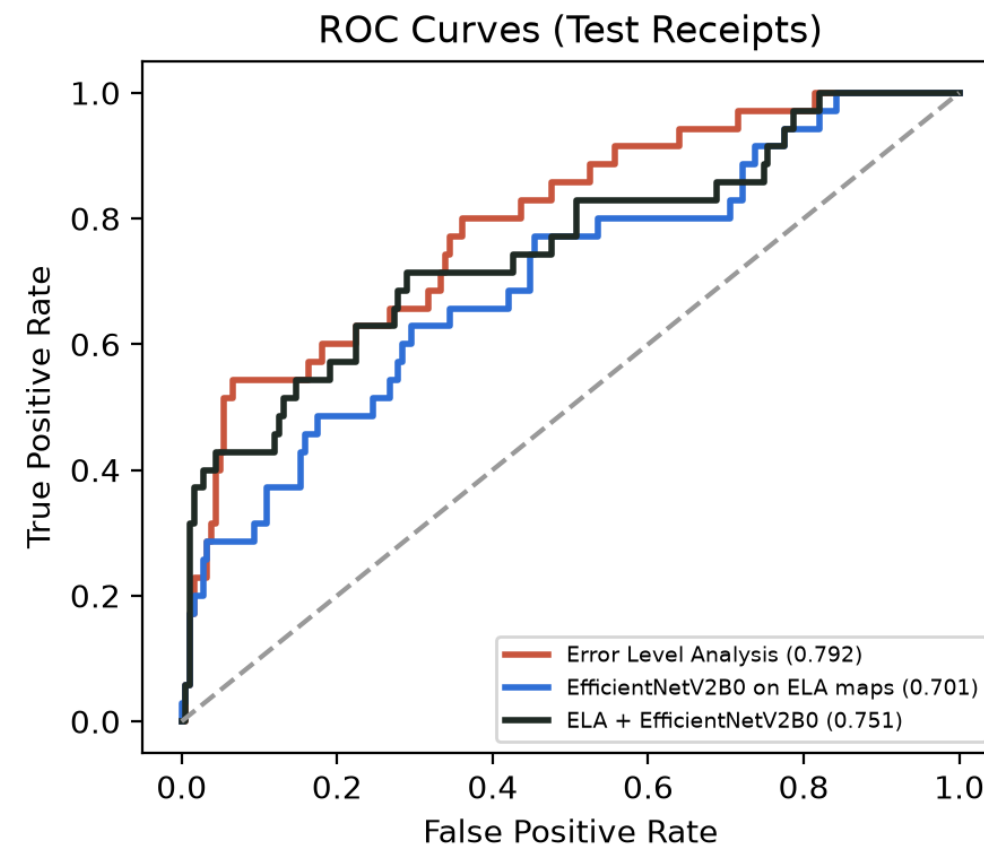
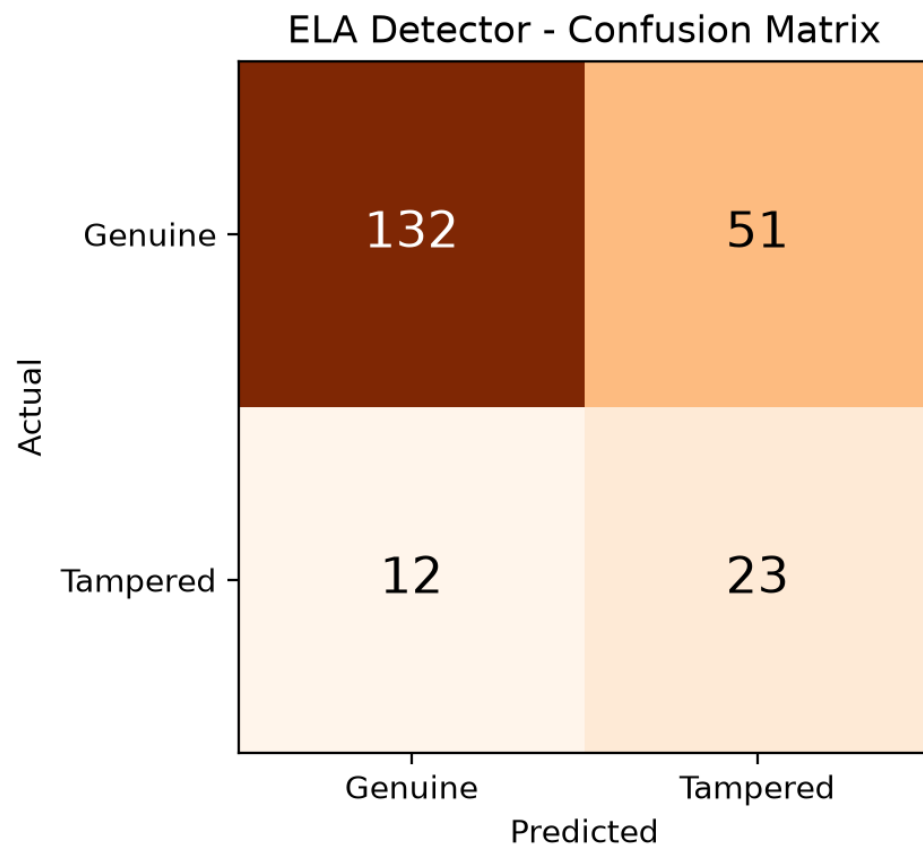
- Windows 10/11, Python 3.10+ (tested with 3.13)
- TensorFlow 2.20 with Keras 3.12 (EfficientNetV2B0)
- Flask 3.1, HTML, CSS, JavaScript
- Pillow, NumPy, PyMuPDF, scikit-learn, matplotlib
- Microsoft Word / PowerPoint (for DOC, DOCX, PPT, PPTX files)

Results on 218 Test Receipts:

- Accuracy: 71.10%
- Precision: 31.08%
- Recall: 65.71%
- F1-score: 42.20%
- ROC-AUC: 79.16% (50% = random guessing)
- Tampered receipts caught: 23 of 35
- Genuine wrongly flagged: 51 of 183



Confusion Matrix and ROC Curve:



From 97.85% (Misleading) to a Working Detector:

- Initial CNN: tiny crops enlarged $3\times$ vs large crops shrunk, so it learned the zoom level (97.85% on patches)
- Tested on whole receipts: 49.5% accuracy, AUC 0.44, i.e. chance level
- Retrained on same-scale pixel patches: still chance level (AUC 0.47, 110 of 183 genuine wrongly flagged)
- Final system with Error Level Analysis: 71.10% accuracy, AUC 79.16%, 51 of 183 genuine flagged
- Lesson: evaluate on whole documents, exactly as the application is used

Screenshots – Tampered Receipt Detected:

FD

FinDocAI

Home

Detect

How It Works

Performance

About

● ANALYSIS COMPLETE

Document Analysis Result

Your financial document has been analyzed using Error Level Analysis, EfficientNetV2B0 and Grad-CAM.

FINAL RESULT

!

Tampered

Tampering score

5.41

An area of the page was compressed differently from the rest of the text.

WHY THIS RESULT

Reasons for the decision

File: X51008099081.png

- One region's compression error differs from the rest of the text by 5.4 standard deviations, which is at or above the decision threshold of 4.0, so the document is reported as tampered.
- 18 of the 542 text regions analysed crossed the threshold. Pasted, re-typed or painted-over areas were saved a different number of times than the rest of the page, so their error level stands out.
- The most unusual region is in the bottom-right part of the page (outlined on the heatmap).
- The EfficientNetV2B0 model trained on ELA maps gives this region a tampering probability of 97.7%; its Grad-CAM map shows which pixels it used.
- File check: The image carries no editing-software information in its metadata.
- Reliability: on 218 test receipts the detector reached 71.1% accuracy and caught 23 of 35 tampered receipts, so treat this result as a screening signal, not proof.

REGIONS ANALYZED

542

Document regions processed

TAMPERING SCORE (Z)

5.41

Most unusual region

DECISION THRESHOLD

4.02

Chosen on validation receipts

FLAGGED REGIONS

18

Regions above the detection threshold

STEP 02

Tampering Heatmap

Where the page is inconsistent

ORIGINAL DOCUMENT



GRAD-CAM (EFFICIENTNETV2B0-ELA)



HOW TO READ THE MAP

● Most unusual

● Somewhat unusual

● Consistent

Screenshots – Digital Files (PDF, Word, PowerPoint):

- Typed Word, PowerPoint and digital PDF files have no scan traces
- The system checks the file metadata instead
- Example: an invoice PDF whose amount was changed and re-saved with iLovePDF
- Result: 3 editing signals found

The screenshot displays the 'Document Analysis Result' page from FinDocAI. The page has a light beige background with a dark header containing the 'FinDocAI' logo and navigation links: Home, Detect, How It Works, Performance, and About. A red status bar at the top indicates 'ANALYSIS COMPLETE'. The main heading is 'Document Analysis Result', followed by a subtext: 'Your financial document has been analyzed using Error Level Analysis, EfficientNetV2B0 and Grad-CAM.' Below this, a yellow box highlights the 'Signs of editing found' section, which includes an information icon, the text 'The file's metadata shows that it was changed after it was created.', and a count of '3' editing signals. The next section, 'WHY THIS RESULT', is titled 'Reasons for the decision' and lists five bullet points explaining the analysis. At the bottom, a preview of 'PAGE 1 (RENDERED)' shows an invoice with the number 'INVOICE INV-2026-0042' and date 'Date: 28/09/2026'. To the right of the preview, the 'HOW IT WAS DECIDED' section lists 'Analysis steps', starting with '1. The file type and structure were checked (Word, PowerPoint or PDF with a text layer).'

FD FinDocAI

Home Detect How It Works Performance About

• ANALYSIS COMPLETE

Document Analysis Result

Your financial document has been analyzed using Error Level Analysis, EfficientNetV2B0 and Grad-CAM.

DIGITAL DOCUMENT · DIGITAL PDF

Signs of editing found Editing signals **3**

The file's metadata shows that it was changed after it was created.

WHY THIS RESULT

Reasons for the decision

File: digital_invoice_edited.pdf

- This is a digital PDF: its pages were produced by software, not scanned or photographed, so there are no scanning or compression traces to analyse. The image-based tampering score is therefore not used for this file.
- Created with: Microsoft Word, iLovePDF.
- The PDF was modified after it was created (created 28 Sep 2026 10:00, modified 29 Sep 2026 18:30).
- The PDF file was saved 2 times (incremental updates). This happens when an existing PDF is changed and saved again.
- The PDF was processed with a PDF editing tool (ilovepdf).
- Signs of editing were found in the file's metadata; compare the document with the original from its issuer.

PAGE 1 (RENDERED)

INVOICE INV-2026-0042
Date: 28/09/2026

HOW IT WAS DECIDED

Analysis steps

1. The file type and structure were checked (Word, PowerPoint or PDF with a text layer).

Conclusion and Future Scope:

FinDoc AI provides a complete, explainable pipeline: images, PDF, Word and PowerPoint files are analysed with Error Level Analysis and an EfficientNetV2B0 model trained on ELA maps, digital files are checked through their metadata, and every result is explained with heatmaps, Grad-CAM and reasons. On whole test receipts it reaches 71.10% accuracy and an AUC of 79.16%; it is an explainable screening aid, live at findoc-ai.vercel.app.

- **Data:** More tampered training data, e.g. automatically generated forgeries
- **Text checks:** OCR-based checks that item prices, totals and tax add up
- **Forensics:** Copy-move and double-JPEG analysis for images compressed many times

THANK YOU